HEIGHT PERCEPTION: LOCATION, MOVEMENT AND IMMERSION IN THE VERTICAL HEMISPHERE

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ABSTRACT

Recent interest in auditory perception incorporating height in acoustic space has developed rapidly over the last few years along with techniques for the production of sound in the vertical hemisphere. While several cinema formats now incorporate overhead loudspeakers, sound artists have been using them for decades, usually in site specific configurations. This paper investigates sound source localization, source movement, spatial impression, immersion and envelopment in 3D soundfields using an array with horizontal and elevated loudspeakers and suggests guidelines for generating perceptually useful 3D environments.

1. INTRODUCTION

Our understanding of auditory perception of sound sources in 3D acoustic space has matured with the psychoacoustic principles involved well defined. On the horizontal plane, inter-aural differences in level (ILD) and time (ITD) provide solid cues for perceptual localization of sound sources. For sources above and below the horizontal plane, spectral cues generated by reflections within the pinna are the strongest cues, particularly where there are no inter-aural differences. Williams [1] identified the complex nature of pinna related ILD and the impact they have on the perception of sound sources at different angles of elevation, see figure 1.

However, compared to inter-aural differences, these elevation spectral cues are weak and vary considerably between listeners, making them impossible to predict and impractical to implement in production. However they do partially account for high frequency sounds being perceived at higher elevations than low frequency sounds when both are reproduced at ear height.

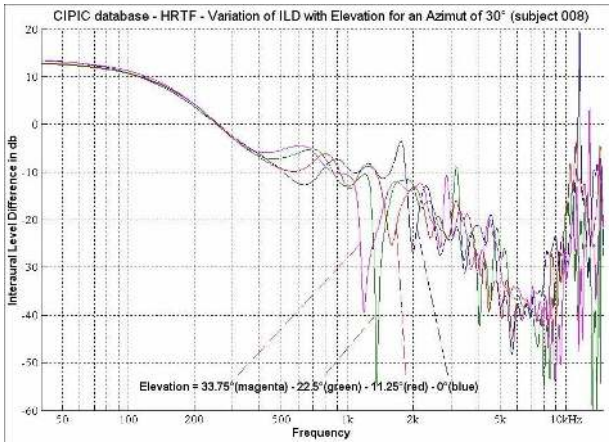


Figure 1: ILD spectral variations for one subject at different elevation angles (after Williams [1]) [2]

Attempts to harness these cues to create virtual height loudspeakers have had very limited success, and in the opinion of the author, never come close to creating the impact of overhead sound sources and immersive environments possible with speakers mounted above the listening position. Consequently, the application of these psychoacoustic principles in loudspeaker arrays incorporating overhead loudspeaker positions requires more investigation.

Another consideration in testing for 3D localization is whether or not to allow participants to move their

head during testing. In research into elevation and azimuth localization reported by Ashby et al [3], accuracy was measured for 3 different cases of self-managed head movement: no movement, forced movement with the head rotating +/-30º azimuth continually, or free movement in any direction. Their results stated that 'For noise programme items, head movements significantly increase elevation localization accuracy' and '…increase azimuth localization accuracy.' They also note 'that pinna cues are less effective than head movement at resolving front-back confusion.'

Head movement (azimuth and elevation) transforms our pinna into a dynamic filter bank of multiple band pass and band reject filters, each with variable centre frequency, variable Q and variable amplitude, thereby endowing an elevated stationary source with perceptual motion and reduced localization blur. As a normal listening environment allows free movement, this position was adopted by the author for this study.

Following on from a previous study by the author of phantom images in the vertical hemisphere [4], this study uses a 3D loudspeaker array to investigate auditory perception of sound source location and movement around and over a listener. The study also investigates spatial impression, immersion and envelopment in 3D reverberant sound fields which may fully envelop the listener, or which appear to be localized to segments of acoustic space.

2. METHODOLOGY

A 3D loudspeaker array is being used for perceptual testing. The Equidome [5] is a 12.2 loudspeaker array with a ring of six speakers on the horizontal plane, six speakers in an elevated ring positioned 60 degrees up from the lower ring and two sub woofers on the floor either side of the listener. One pair of speakers in each ring is on the frontal plane, (+/- 90º) and all others are at 60º angles from that pair, ie. +/-30º and +/-150º. All speakers used are Bose cubes which have been equalized to be consistent in timbre and amplitude, and are spaced on a 1.5 metre radius from the listening position in a small room with substantial wall and ceiling treatment to minimize acoustic anomalies, see figure 2. All speakers were visible to the participants which is consistent with a normal home listening environment.

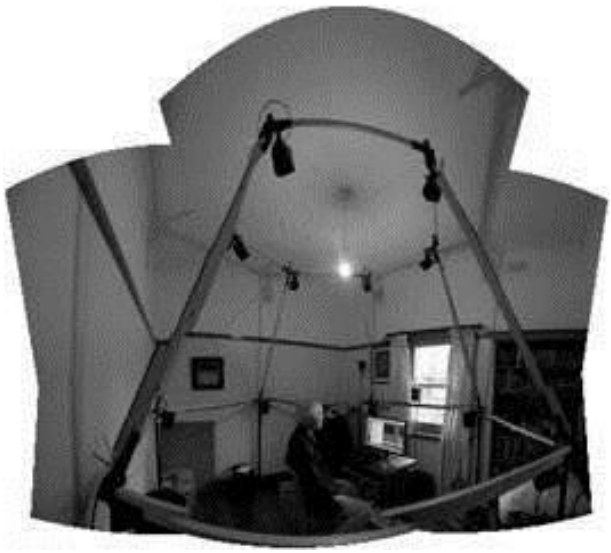


Figure 2: The Equidome

Nuendo software allows placement and automated movement of multiple, simultaneous sound sources anywhere throughout the array and Waves 360 convolution reverb allows the creation of multiple early and late reflections, each of which may be placed individually. Perceptual testing using a small group of expert listeners and controlled double-blind procedures has been completed and a quantitative and qualitative analysis has been undertaken.

3. RESULTS

3.1. Location

The first stage of testing focussed on static sound source locations around and over the listeners using male speech and a vibraslap percussion sample. Results indicate that differences in the accuracy of the perceived location, spectral changes and localization blur depend on whether the source is located totally within one speaker, hereafter called in-speaker, as a phantom image created by amplitude panning between two speakers, or as an image created through amplitude panning between four speakers.

When the sound sources are located in-speaker, it was clear that all participants perceived the sounds at that speaker position with no localization blur and with a spectral character which became the benchmark for subsequent comparison. For locations on the horizontal plane, limited head movement was observed for participants to be confident in selecting a location. When the in-speaker location was on the elevated plane, participants moved their heads using small turning and tilting motions, and successfully located the sources at the front, side or rear speaker positions.

Using amplitude panning to locate sources halfway between two speaker positions on the horizontal plane with the usual 3dB panning law, the results were very accurate for front centre and rear centre, with slightly reduced accuracy and a small localization blur for side locations, for example left front to left side or left side to left rear. This is consistent with knowledge of phantom images for a 60º stereo speaker pair where inter-aural time and level differences work their magic but become less effective at lateral locations. It is noted that speaker locations on the frontal plane at +/- 90º are very important for localization around the listener when combined with a rear pair at +/-150º and overcome poor horizontal localization when using the standard 5.1 arrangement of +/-110º.

When the sources are amplitude panned halfway between a horizontal and elevated speaker pair, for example horizontal front left to elevated front left (60º elevated), azimuth angle estimation was very good but elevation angle results were inconsistent with a tendency to localize at the elevated speaker rather than at a mid-point. Head movement was used by all participants and observed localization blur was greater, while some participants also perceived a spectral change.

When sources were panned between elevated only pairs, elevation perception was consistently accurate but azimuth angle estimation was less accurate with some front-rear confusion observed and localization blur greater than for the other two speaker phantom locations. The most accurate localization was for elevated left side and elevated right side speakers which are on the frontal plane, where inter-aural differences combine more effectively with elevation spectral changes.

The final test results were obtained when the phantom image was generated using amplitude panning between four speakers, for example between horizontal front left and right and elevated front left and right, including directly overhead where there is no speaker. In these cases, the phantom location was perceived with good azimuth angle estimation with a small localization blur increasing away from front centre and rear centre. Elevation angle perception again tended to be consistently located at a higher angle and was always above the horizontal plane. Head movement assisted in locating the sources and the top centre position was accurately located by all participants. Spectral changes were again noted by several participants.

Location Discussion

Previous studies of average angular localization errors, notably in Best at al [6] and Hammershoi [7], were based on studies conducted with a moving speaker or an array with a very large number of speakers. This removed any difficulties posed by phantom image generation but there were inaccuracies caused by methodologies used for indicating the location of a perceived sound source, with limitations due to head movement tracking, nose mounted pointing devices and other techniques. It was noted that the angular error increased as sources move away from the front, and up and over the top.

From the study presented here, using an array with a limited number of speakers, in-speaker locations were accurately located in both the horizontal and elevated sectors of the vertical hemisphere. When the sound source location was generated using amplitude panning between 2 speakers on the horizontal plane, the phantom image was perceived with good azimuth accuracy and little blur. A phantom image generated on the elevated plane was perceived with good elevation angle estimation but poor azimuth angle estimation, while on the vertical planes between 2 speakers, the accuracy was good for azimuth but poorer for elevation, though the sound sources were always perceived to be elevated. Four speaker phantom images generally had good azimuth perception but greater blur for elevation, though again they were always perceived to be elevated. In particular, for elevated sources on the frontal plane including directly overhead, the coherence of elevation spectral cues and inter-aural differences results in good accuracy with reduced localization blur.

	Azimuth accuracy	Elevation accuracy	Localization blur
in-speaker location	excellent	excellent	none
phantom image: **2 speaker horizontal only**	very good	n/a	low
phantom image: **2 speaker elevated only**	some front/rear confusion	very good	medium
phantom image: **2 speaker vertical plane**	very good	ok, tends to elevated positions	medium
phantom image: **4 speaker horizontal and elevated**	good	ok, tends to elevated positions	medium, worse at rear elevated postions

Table 1 3D localization

3.2. Movement

For the second test, a stereo sample of broadband noise was created using inter-channel amplitude panning using the auto-pan function within Adobe Audition with the 3dB panning law. The noise source panned from left

to right and back to the left over 4 seconds and for testing, the left channel is allocated to one speaker and the right to a second speaker. Trajectories have been mapped around and over the listeners with an angular range of 60º, and have included horizontal only paths, elevated only paths, vertical paths from a horizontal speaker to an elevated speaker, across the top on the frontal plane and the median plane, and diagonal paths including horizontal front left to elevated front right.

Results indicate a clear sense of movement along the path in all sectors of 3D acoustic space, but with the start and finish points subject to similar errors to the static locations. When the path crosses the median plane, for example from horizontal front left to horizontal front right, there was a clear sense for all participants of a point source moving between the 2 speakers, consistent with our understanding of stereo phantom images and inter-aural differences. This result was also achieved for all participants for horizontal rear left to right which is to be expected. It was also interesting to note that where the path crossed the median plane at elevated positions, for example from elevated left side to elevated right side, there was 100% accuracy in identifying the path of movement across the top, reflecting the coherence of inter-aural cues and elevation spectral cues.

When the paths were along the horizontal plane only between front and side or side and rear locations, with the start and finish point in-speaker, perception of the path was again very accurate for all participants. However, when the path was along the elevated plane only, there was confusion between front and rear in-speaker locations, which were partially resolved by head turning, but indicated a lack of confidence in locations where spectral cues are weak.

Paths which started at horizontal speakers and moved vertically to elevated speakers were consistently identified with accurate azimuth and a clear sense of movement up and down. Of particular interest with this group of paths was the observation by several participants that the noise source no longer appeared to be a point source moving along a path but appeared now to be a noise source fading in and out at each speaker, with significant blur at the mid-point of the transition. This observation is consistent with no inter-aural change between speaker locations and only spectral cues providing localization information.

The final group of paths tested were diagonal trajectories, for example between horizontal front left and elevated front right. Where these paths crossed the median plane, there was 100% accuracy in identifying the path both for azimuth and elevation, indicating solid inter-aural cues and spectral cues. When the path was around the listener, for example horizontal front left to elevated side left, there was again very good accuracy in both azimuth and elevation perception. However, when the path was wholly at the rear but did not cross the median plane, and also when the path was along the median plane, there was some confusion with the exact starting and finishing locations, consistent with previous location results. There was, nevertheless, a clear perception of movement.

Movement Discussion

The perception of movement for a broadband noise source is clear throughout the vertical hemisphere when the source is panned between 2 in-speaker locations. When the sound source crosses the median plane along the path, there is very accurate perception of the azimuth angle. The elevation angle is also very clear everywhere except rear positions where there is increased localization blur. As identified by Williams [1], diagonal paths between horizontal and elevated speakers are also clear for most paths, though again there is increased blur for start and end locations at the rear. Vertical paths at all azimuth angles tend to exhibit a transition which is not as smooth between the two end points, as elevations cues are not as accurate as inter-aural cues, and the sound source tends to become two isolated sources fading in and out at each point.

It should be clearly understood that this accuracy in localization and movement directly overhead is only possible with speakers elevated 60º from the horizontal plane. If they are below this level, the gap across the top is too large for meaningful phantom locations or movement, though it may be suitable for enhanced 3D reverberant soundfields.

	Azimuth angle perception	Elevation angle perception	Localization blur
horizontal only path across **median plane**	excellent	n/a	none
elevated only path across **median plane**	excellent	very good	low
diagonal path across **median plane**	excellent	very good	low
horizontal only path along **sides**	good	n/a	low
elevated only path along **sides**	ok, some front/rear confusion	good	medium
diagonal path at **sides**	good	good	medium
median plane path	ok	ok	medium, worse at rear
vertical path	very good	ok, tends to isolated images top and bottom	medium, worse at middle point

Table 2 3D movement perception

3.3. Spatial Impression, Immersion and Envelopment

Early and late reflection patterns are generated in virtually all acoustic environments and we perceive them as integral parts of the listening experience. There are two broad categories of reverberant soundfields: those generated by static sound sources in a reverberant space, for example a performance on stage in a concert hall, and urban or country environments where a reverberant soundfield is generated in a smaller localizable sector in acoustic space, and where multiple, different reverberant soundfields may exist simultaneously. There are interesting questions raised about both categories when we consider creating or reproducing similar soundfields, which include the significance of height in the perception of acoustic space.

As Theile and Wittek [8] suggested, '…the perception of distances and depth mostly depends on early reflections,…(and) perception is particularly stable when the reflections come in from the original directions of the upper half space.' Control of early and late reflections and the directions from which they arrive at the listener is of critical importance in generating realistic immersive environments within a 3D speaker array.

The final stage of testing has investigated perception of the reverberant space generated with different balances between horizontal and elevated reflection patterns for a fully immersive and enveloping static environment, and for a reverberant space created around a static source in a small localizable sector in acoustic space. There has also been an investigation into the balance of early reflections between the horizontal plane and the elevated plane for the generation of clear spatial impression, and the balance of late reflections between horizontal and elevated for immersion and envelopment.

The sound sample used for this sequence of tests was a 4 second acoustic guitar phrase recorded in anechoic conditions which was convolved in the Waves 360 reverb software to generate a series of 4 channel early reflections (ER) and separate 4 channel late reflections (LR), a technique inspired by the Spacebuilder concept [9]. Each of the ER and LR components, along with the mono original sound were individually placed around the listeners. For example, the guitar was panned halfway between horizontal front left and horizontal side left speakers, and the early and late reflections were panned in-speaker at horizontal front left and side left and elevated front left and side left. This created an acoustic space located in only one sector of the vertical hemisphere, and with appropriate changes to the locations, any sector could be activated with the direct sound source and reflections. Using software automation, the direct, ER and LR components can be moved around the listener creating a sense of movement of a reverberant soundfield in acoustic space.

Results indicate that all participants found the acoustic space generated was very clear and could easily identify in which sector it was located. The participants were also able to clearly perceive the added sense of immersion generated by the elevated speakers. Two special cases were also tested: all components including the direct sound were panned to elevated speakers only, or to horizontal speakers only. When all component sources were elevated, this was clearly identified as totally overhead and separate from the horizontal plane. When all the components were horizontal only, participants felt fully enveloped in the sound.

As a further test of the capacity to perceive individual sectors of reverberation, a 2 minute percussion piece was played to participants. This piece had been generated by recording six separate percussion instruments in near anechoic conditions and generating distinctly different ER and LR 4 channel reverberation patterns for each part. These parts and their corresponding reverb were panned to each of the six sectors around the participants. All participants could clearly perceive a different spatial impression and reverb identity for each part and felt immersed in the overall soundfield.

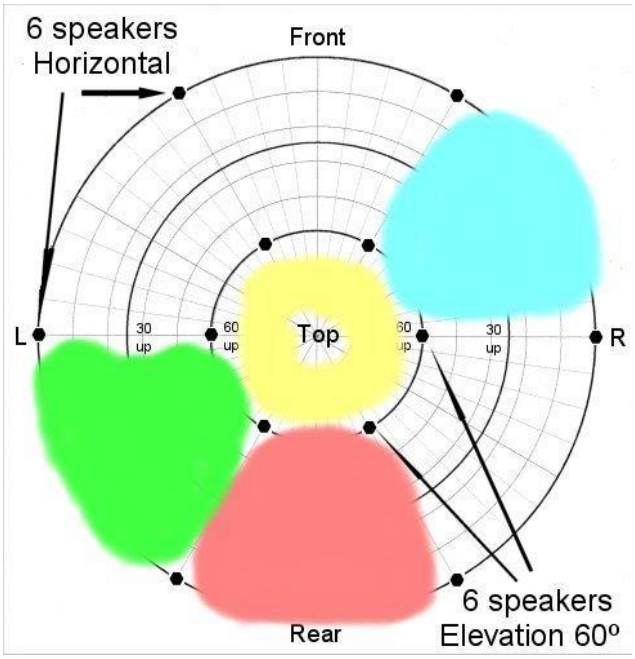


Figure 3: 4 instances of unique spatial impression and reverb identity in 4 separate sectors

These results confirm that it is possible to generate localizable soundfields, each with a different spatial impression and reverb identity incorporating a sense of height, in different sectors around and over a listener and that multiple instances of acoustic space may be perceived simultaneously in separate sectors of the total soundfield around a listener, see figure 3.

4. DISCUSSION AND CONCLUSIONS

This paper has investigated whether it is possible to generate perceptually accurate 3D soundfields with sounds located around and above a listener. An array of six speakers on the horizontal plane and 6 speakers on an elevated plane was used to generate stationary and moving sound sources and to generate reverberation.

When stationary sound sources are panned to in-speaker locations, they can be located with excellent accuracy including behind and above the listener. However, when the sound source is created using phantom images, there is variation in localization accuracy. For phantom images on the horizontal plane, accuracy is very good, but for phantom images on the elevated plane, elevation perception is good but azimuth perception suffers some front/rear confusion which head turning is unable to fully resolve. When phantom images are generated on vertical planes between speakers, azimuth localization is good but elevation perception tends to the elevated speaker position.

Movement perception is very good in all areas of the array but accurate localization of start and finish positions depends on the trajectory. When the path crossed the median plane and inter-aural differences are produced, there is very good accuracy in locating the path, including across the front and back, over the top and diagonally up or down. However, similarly to stationary source perception, it is difficult to create accurate localization for sounds moving vertically between speaker layers even though a sense of movement is still strong.

Using convolution reverb software to generate multiple early and late reflections, which are then placed around and over the listener, allows the creation of 3D soundfields which can accurately emulate the perception of a concert hall, fully immersing the listener with strong spatial impression and enveloping reverberation. If, however, the direct sound, early and late reflections are individually panned to only one part of the array, a localized soundfield with distinct spatial impression and reverb identity can be created in a sector of the soundfield. This can be very useful for a sound artist creating unique acoustic spaces around a listener, or it can enhance the mix for a movie.

Chris Scarabosio, sound mixer for the film ***Elysium***, when talking about mixing for the new cinema formats of Dolby Atmos [11] and Auro 3D [12], reported in Audio Technology magazine [10] '…using reverb not to create more ambience but to localize the ambience.' He was referring to one scene where he created an ambient space around the music mix in one sector of 3D acoustic space, while using the full 3D speaker array of both Atmos and Auro 3D for other components of the sound design.

This study has shown that a 3D loudspeaker array with speakers elevated 60º from the horizontal plane can generate perceptually useful localization of stationary and moving sound sources around and above listeners. The array can also produce 3D soundfields with clear spatial impression, immersion and envelopment in a single acoustic space or produce multiple, simultaneous acoustic spaces around and above the listener, each of which may have unique spatial impression and immersive identity. The use of a 3D loudspeaker array offers exciting opportunities to sound designers and sound artists to explore the creative potential of immersive aural environments.

5. REFERENCES

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